

Validation

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1. Introduction

The purpose of this document is to verify and validate the final implementation of the Secure Notepad Web Application.

Validation ensures that the application fulfills its intended purpose — offering users a secure way to store and manage notes through encrypted communication. Verification confirms that each functional requirement, as defined in the **Analysis** and **Design** documents, has been correctly implemented and operates as intended.

This document includes the applied testing methodology, detailed feature tests, environment specifications, and feedback from stakeholders to demonstrate that the final product meets its technical goals.

2. Validation Objectives

The objectives of the validation phase are:

- To ensure that the implemented system meets the **functional** and **non-functional** requirements.
- To verify that HTTPS encryption and secure communication are correctly configured.
- To confirm that the application's backend and frontend behave as expected.
- To validate database interaction, initialization, and data persistence.
- To collect stakeholder feedback for performance, usability, and reliability assessment.

3. Test Plan and Methodology

Testing was performed manually and through observation-based verification.

Each test is defined by:

- **Test ID**
- **Objective**
- **Precondition**
- **Expected Result**
- **Actual Result**
- **Status** (Pass / Fail / Needs Improvement)

Testing focused on functional correctness, backend logic, database operations, and secure hosting validation.

4. Test Environment

Hardware:

- Lenovo ThinkPad T480 (Development and testing system)
- Raspberry Pi (Hosting platform)

Software:

- Operating System: Linux

- Web Framework: Flask (Python 3.x)
- Database: SQLite
- Security: Certbot (for HTTPS certificates)
- Web Browser: Firefox

5. Feature Validation

5.1 HTTPS and Security Validation

Test ID	Objective	Precognition	Expected Result	Actual Result	Status
Security-1	Verify that HTTPS is properly configured and encryption is active.	Raspberry Pi server is running with Certbot certificates installed.	Website loads with HTTPS; browser shows “Secure Connection” .	HTTPS is active and verified through browser padlock and SSL certificate information.	PASS

Screenshots of Certificate Details:



Figure 1. Proof of padlock next to domain proving secure connection

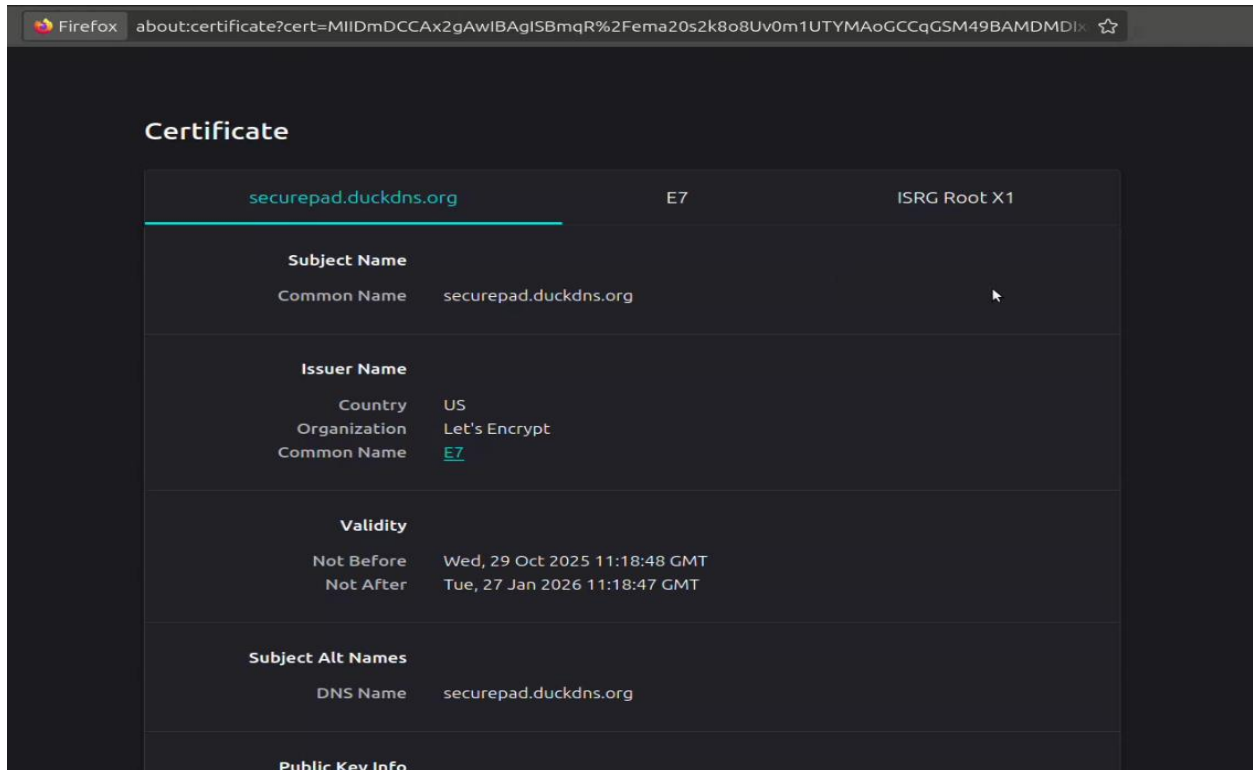


Figure 2. Proof of certificates used for secure connection

5.2 Hosting Accessibility Validation

Test ID	Objective	Precognition	Expected Result	Actual Result	Status
Hosting-1	Confirm that the web application is accessible to the public network.	Server is powered and connected to local network with open ports.	Application reachable via public IP or domain from an external network / VPN.	Application accessible; verified through external device connection.	PASS

5.3 User Account Management Validation

Test ID	Objective	Expected Result	Actual Result	Status
Auth-1	Ensure user registration stores data securely.	User credentials hashed using Werkzeug security and stored correctly.	Hashing verified in database; passwords unreadable and salted.	PASS

Auth-2	Validate login/logout functionality.	Login with valid credentials succeeds; invalid login fails.	Both verified. Invalid attempts handled gracefully.	PASS
Auth-3	Verify account deletion functionality.	User account removed from database after confirmation.	Account deleted successfully; database updated.	PASS

5.4 Notes Creation and Management Validation

Test ID	Objective	Expected Result	Actual Result	Status
Notes-1	Confirm creation of new notes.	Note saved under user's account and retrievable after refresh.	Verified; data persists in SQLite.	PASS
Notes-2	Validate note editing and updating.	Changes reflected immediately after save.	Verified through frontend and database logs.	PASS
Notes-3	Check encryption and decryption workflow.	Notes encrypted using substitution cipher; user must provide key for decryption.	Encryption/decryption successful with correct key.	PASS

5.5 Database Initialization Validation

Test ID	Objective	Expected Result	Actual Result	Status
DB-1	Check if database initializes automatically on first launch.	Application creates database structure on start.	Requires manual initialization.	FAIL

6. Stakeholder Feedback

Feedback collected during development and after testing:

- **Ceryl:** Guided the setup of Nginx and server-side configuration; confirmed successful hosting and HTTPS setup.
- **Xuemei:** Recommended HTTPS implementation and advised transition from desktop to web app for broader learning scope.
- **Peers:** Provided usability feedback on registration and note management, reporting that the interface is intuitive and reliable.

7. Conclusion

The validation process confirmed that the Secure Notepad Web Application successfully fulfills the core objectives established in the analysis phase. Each validation test directly supports a specific educational or technical goal defined earlier:

- **Secure Communication (Goal: HTTPS Configuration)** - Verified through successful HTTPS setup on the Raspberry Pi using Certbot, ensuring encrypted data transmission between client and server.
- **Data Security and Authentication (Goal: Encryption and Safe Storage)** - User credentials are hashed with Werkzeug security, and encrypted notes can only be decrypted with the correct key, validating the encryption and authentication requirements.
- **System Accessibility (Goal: Reliable Hosting and Deployment)** - Hosting was confirmed functional and publicly accessible, aligning with the deployment and networking learning objectives.
- **Usability and Functionality (Goal: User Experience and Core Features)** - Note creation, editing, and account management were validated as stable and intuitive, demonstrating that functional and user-oriented goals were achieved.
- **System Initialization (Goal: Automation and Reliability)** - The only partial failure involved the database initialization process, which still requires manual setup. This identifies a clear target for future improvement to enhance automation and deployment robustness.

Through this validation, the project demonstrates that the implemented solution not only meets its functional requirements but also reinforces the main educational goals - applying encryption, authentication, and secure hosting in a real-world context.